

ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH



SESSION:-2025-26

EVALUATION OF INTERNSHIP REPORT

CODE:CY- BT107

**Submitted To:
Prof. Shruti Lashkari**

**Submitted By:
Name: Vagisha Goyal
Enrollment No: 0827CY241071**



**Department of Computer Science and Engineering
(Cyber Security)**

AITR, Indore,

ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH, INDORE



Department of Computer Science and Engineering (Cyber Security)

Certificate

Certified that the training work entitled “C / C++ / Java / Python / MERN Stack” is a bona fide work carried out by “Vagisha Goyal” after the completion of the Third Semester, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science Engineering (Cyber Security), under the guidance of Prof. Shruti Lashkari, Acropolis Institute of Technology and Research, during the academic year 2025–2026.

Name and Sign of Training Coordinator

ACROPOLIS INSTITUTE OF TECHNOLOGY & RESEARCH, INDORE



Department of Computer Science and Engineering
(Cyber Security)

ACKNOWLEDGEMENT

I would like to acknowledge the contributions of the following people without whose help and guidance this report would not have been completed. I acknowledge the counsel and support of our training coordinator Prof. Shruti Lashkari, CSE(CY) Department, with respect and gratitude, whose expertise, guidance, support, encouragement, and enthusiasm has made this report possible. Their feedback vastly improved the quality of this report and provided an enthralling experience. I am indeed proud and fortunate to be supported by him/her. I am also thankful to Dr. Shilpa Bhalerao, H.O.D of Computer Science Information Technology Department, for her constant encouragement, valuable suggestions and moral support and blessings. Although it is not possible to name individually, I shall ever remain indebted to the faculty members of CSE(CY) Department, for their persistent support and cooperation extended during this work.

Student Name :- Vagisha Goyal

Student Enrollment No. :- 0827CY241071

ACROPOLIS INSTITUTE OF TECHNOLOGY & RESEARCH, INDORE

INDEX

S.No CONTENTS

- | | |
|---|--|
| 1 | Assignment problems /Practice Questions with output[gdb] (15Marks) |
| 2 | Proof of No. of Problems solved in Codechef/HackerRank/LeetCode / other coding Platforms (Progress Report) (20 Marks) |
| 3 | Project Report and screenshot and code (20 marks) |
| 4 | MERN STACK course by IIT Ropar certificate by IIT Ropar (Certificate and report) (20 Marks) |
| 5 | IBM Skill boost course /training Certificate (10 marks) (Certificate and report) |
| 6 | NPTEL/ GLOBAL Certification / Hackathons/ Coding Competition/ (15 marks) |

Sr.No	Programs Title
1	<u>EOI Given a string, print only the characters that appear at odd positions (index 1, 3, 5...).</u>
2	<u>Find the Sum of Digits Without Using a Loop Problem: Take a number as input and print the sum of its digits using recursion only.</u>
3	<u>First Non-Repeating Character Problem: Given a string, find the first character that does NOT repeat.</u>
4	<u>Given a string, print only the characters that appear at odd positions (index 1, 3, 5...).</u>
5	<u>Replace All Zeros with Ones Problem: Input: 102030 Output: 112131</u>
6	<u>Check if a character is a vowel or consonant.</u>
7	<u>Print even numbers from 1 to 50.(for loop)</u>
8	<u>Calculate the sum of the first n natural numbers</u>
9	<u>Print the multiplication table of any number</u>
10	<u>Print the reverse of a given string.</u>
11	<u>Print each character of a string separately</u>
12	<u>Draw Pattern shown in description</u>
13	<u>Print Fibonacci series up to n terms</u>
14	<u>Check if a year is a leap year.</u>
15	<u>Check if a number is divisible by 5 and 11 or not.</u>
16	<u>Check if a person is eligible to vote (age >= 18).</u>
17	<u>Check if a given alphabet is uppercase or lowercase.</u>
18	<u>Check if a character is a vowel or consonant.</u>
19	<u>Untitled AssignmentWrite a menu-driven Python program using functions to perform different operations on numbers.</u>
20	
21	
22	
23	
24	

Practice Questions with **output[gdb]**

Submitted version ▾

Re-Submit

   Run ▾  Debug  Stop  {} Beautify  ▾  Language C++ ▾

main.c

```
1 #include <stdio.h>
2
3 int main() {
4     char str[100];
5     scanf("%s", str);
6
7     for (int i = 1; str[i] != '\0'; i += 2)
8         printf("%c", str[i]);
9
10    return 0;
11 }
12
```

     input

45123f
52f

...Program finished with exit code 0
Press ENTER to exit console.

Submitted version ▾

Re-Submit

Run ▾

Debug

Stop

{ } Beautify



Language C++ ▾

main.c

```
1 #include <stdio.h>
2
3 int main() {
4     char str[100];
5     scanf("%s", str);
6
7     for (int i = 1; str[i] != '\0'; i += 2)
8         printf("%c", str[i]);
9
10    return 0;
11 }
12
```

input

45123f

52f

...Program finished with exit code 0
Press ENTER to exit console.

Run Debug Stop Beautify Language C++

main.cpp

```
1
2 #include <iostream>
3 #include <unordered_map>
4 using namespace std;
5 int main() {
6     string str;
7     cin >> str;
8
9     unordered_map<char, int> freq;
10
11
12     for (char c : str) {
13         freq[c]++;
14     }
15
16
17     for (char c : str) {
18         if (freq[c] == 1) {
19             cout << c;
20             return 0;
21         }
22     }
23
24     cout << "No non-repeating character";
25     return 0;
```

input

qwerr
q

...Program finished with exit code 0
Press ENTER to exit console.

Run Debug Stop Beautify Language C++

main.cpp

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     string num;
6     cin >> num;
7
8     for (char &c : num) {
9         if (c == '0')
10             c = '1';
11     }
12
13     cout << num;
14     return 0;
15 }
16
```

input

```
254
254
...Program finished with exit code 0
Press ENTER to exit console.
```

Draft version ▾

Submitted version



Run ▾

Debug

Stop

Save

Beautify

submit



Language Python 3 ▾

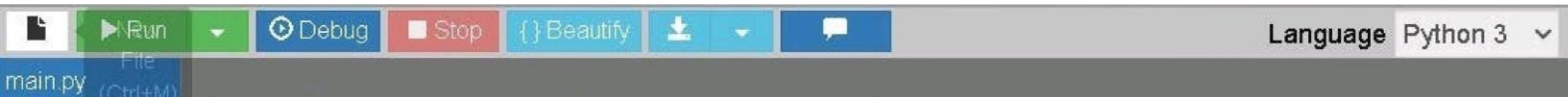
main.py

```
1 for i in range(2, 51, 2):  
2     print(i)  
3
```

input

44
46
48
50

...Program finished with exit code 0
Press ENTER to exit console.



```
1 input_char = input("Enter a single character: ")
2
3 if len(input_char) == 1 and input_char.isalpha():
4     char_lower = input_char.lower()
5     vowels = "aeiou"
6     if char_lower in vowels:
7         print(f"The character '{input_char}' is a vowel.")
8     else:
9         print(f"The character '{input_char}' is a consonant.")
10 else:
11     print("Invalid input. Please enter a single alphabet character.")
12
```

input

Enter a single character: l
The character 'l' is a consonant.

...Program finished with exit code 0
Press ENTER to exit console.

Run Debug Stop {} Beautify Language Python 3

main.py

```
1
2 n = int(input("Enter a number: "))
3 sum_n = 0
4 for i in range(1, n + 1):
5     sum_n += i
6 print("Sum of first", n, "natural numbers is:", sum_n)
7
```

input

Enter a number: 1298
Sum of first 1298 natural numbers is: 843051

..Program finished with exit code 0
press ENTER to exit console.

Run Debug Stop Beautify Language Python 3

main.py

```
1 n = int(input("Enter a number: "))
2 for i in range(1, 11):
3     print(n, "x", i, "=", n * i)
4
```

input

```
423 x 7 = 2961
423 x 8 = 3384
423 x 9 = 3807
423 x 10 = 4230
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

Run Debug Stop Save Beautify submit Language Python 3

main.py

```
1 s = input("Enter a string: ")
2 print(s[::-1])
3
```

input

Enter a string: darsh
hsrad

...Program finished with exit code 0
Press ENTER to exit console.

 Run  Debug  Stop Save { } Beautify submit 

Language Python 3 

main.py

```
1 s = input("ENTER STRING ")
2 for ch in s:
3     print(ch)
```

     input

ENTER STRING darsh

d
a
r
s
h

...Program finished with exit code 0
Press ENTER to exit console.

Run Debug Stop Save { } Beautify submit Language Python 3

main.py

```
1 n=int(input("ENTER NO. OF ROWS"))
2 for i in range(0,n):
3     print("  *(n-i-1)+*"  *(i+1))
4     print()
```

input

ENTER NO. OF ROWS4

*

* *

* * *

* * * *

...Program finished with exit code 0

Press ENTER to exit console.

Run

Debug

Stop

{ }

Beautify

Language Python 3

main.py

```
1 n = int(input("ENTER NUMBER "))
2 a, b = 0, 1
3
4 for _ in range(n):
5     print(a, end=" ")
6     a, b = b, a + b
```

input

ENTER NUMBER 5
0 1 1 2 3
...Program finished with exit code 0
Press ENTER to exit console.

Run Debug Stop { } Beautify Language Python 3

```
main.py
1 year = int(input("Enter a year: "))
2
3 if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
4     print("Leap year")
5 else:
6     print("Not a leap year")
```

input

Enter a year: 2020
Leap year

...Program finished with exit code 0
Press ENTER to exit console.

```
main.py
1 num = int(input("Enter a number: "))
2
3 if num % 5 == 0 and num % 11 == 0:
4     print("Divisible by 5 and 11")
5 else:
6     print("Not divisible by 5 and 11")
```

input

Enter a number: 55
Divisible by 5 and 11

...Program finished with exit code 0
Press ENTER to exit console.

Draft version ▾

Submitted version



Run ▾

Debug

Stop

Save

{ } Beautify

submit



Language Python 3 ▾

main.py

```
1 age = int(input("Enter your age: "))
2
3 if age >= 18:
4     print("Eligible to vote")
5 else:
6     print("Not eligible to vote")
```



input

```
Enter your age: 18
Eligible to vote
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

Submitted version ▾

Re-Submit



Run ▾

Debug

Stop

{ } Beautify



Language Python 3 ▾

main.py

```
1 ch = input("Enter an alphabet: ")
2
3 if ch.isupper():
4     print("Uppercase")
5 else:
6     print("Lowercase")
7
```

input

Enter an alphabet: d
Lowercase

...Program finished with exit code 0
Press ENTER to exit console.

Run Debug Stop { } Beautify Language Python 3

main.py

```
1 ch = input("Enter a character: ").lower()
2
3 if ch in 'aeiou':
4     print("Vowel")
5
6 else:
7     print("Consonant")
```

input

Enter a character: h
Consonant

...Program finished with exit code 0
Press ENTER to exit console.

Draft version ▾

Submitted version

Run ▾

Debug

Stop

Save

{ } Beautify

submit

Language Python 3 ▾

main.py

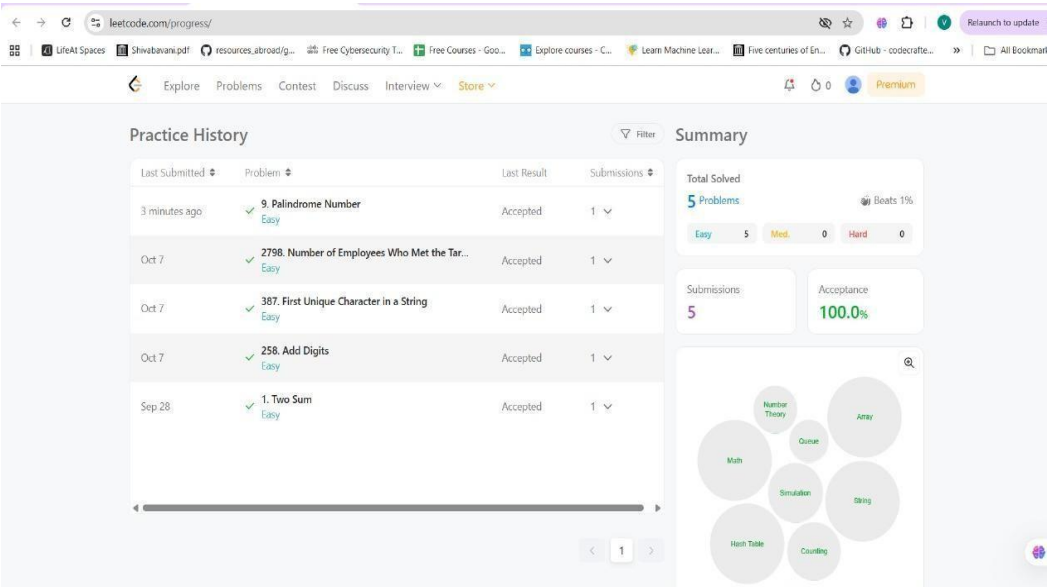
```
1 def add(a, b):
2     return a + b
3
4
5 def subtract(a, b):
6     return a - b
7
8
9 def multiply(a, b):
10    return a * b
11
12
```

input

```
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 1
Enter first number: 3
Enter second number: 4
Result: 7.0
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 
```


PROOF OF NO. OF PROBLEMS
SOLVED IN
CODING PLATFORMS

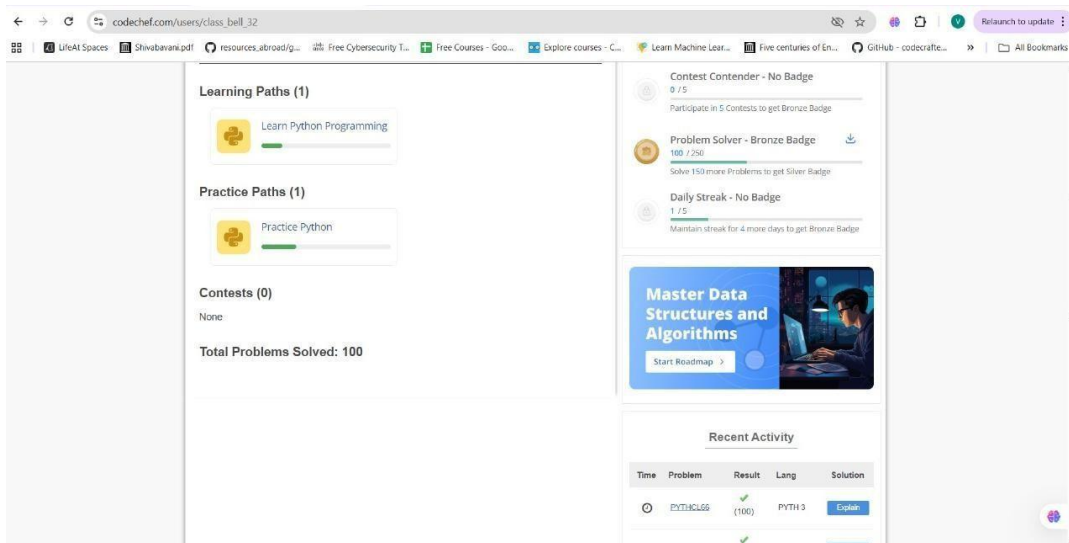
Leetcode:



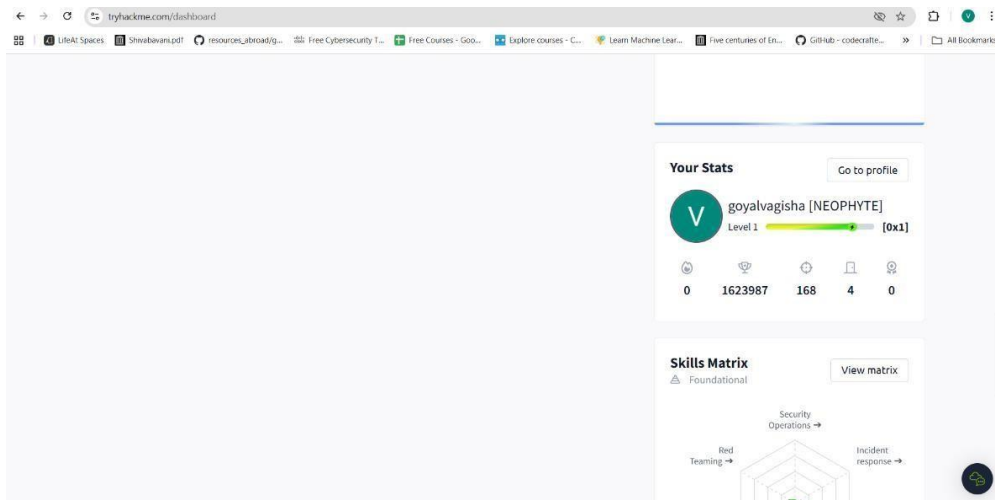
HackerRank:

Problem	Language	Time	Result	Score	Action
Python: Division	python3	1 minute	Accepted	10.0	<button>View results</button>
Python If-Else	python3	2 minutes	Accepted	10.0	<button>View results</button>
Python If-Else	python3	3 minutes	Wrong Answer	8.33	<button>View results</button>
Say "Hello, World!" With Python	python3	4 minutes	Accepted	5.0	<button>View results</button>
Arithmetic Operators	python3	5 minutes	Accepted	10.0	<button>View results</button>
Python If-Else	python3	7 minutes	Wrong Answer	8.33	<button>View results</button>
Python If-Else	python3	8 minutes	Wrong Answer	8.33	<button>View results</button>
Python If-Else	python3	9 minutes	Wrong Answer	8.33	<button>View results</button>

CodeChef:



Try Hack Me:



PROJECT REPORT

Title of the Project: TextShield

Subject Name: EOI

Student Name: Urvashi Misal

Roll Number: 0827CY241070

Student Name: Vagisha Goyal

Roll Number: 0827CY241071

Student Name: Vaidika Rai

Roll Number: 0827CY241072

Student Name: Vanshika Choudary

Roll Number: 0827CY241073

Class & Section: CY Second Year 3rd sem

Academic Year: 2025–26

Submitted To: Prof. Shruti Lashkari

- **Introduction**

With the rapid growth of digital communication, protecting sensitive information has become essential. Encryption is a technique used to convert readable data (plaintext) into an unreadable format (ciphertext) to prevent unauthorized access. This project demonstrates a simple encryption and decryption system using Python, helping beginners understand basic cryptography concepts and logic implementation.

- **Problem Statement**

In many basic systems, data such as messages or passwords are stored or transmitted in plain text, making them vulnerable to misuse and data breaches. There is a need for a simple mechanism that can secure data by encrypting it and later retrieving the original information through decryption.

• **Proposed System / Solution**

The proposed system uses a simple encryption algorithm where characters in a message are transformed using a predefined logic (such as character shifting or substitution).

- **Encryption** converts the original message into an encoded format using a key.
- **Decryption** reverses the process using the same key to retrieve the original message.

The system is user-friendly, menu-driven, and suitable for understanding core encryption concepts.

• **Tools and Technologies Used**

Tools and Technologies Used

- **Programming Language:** Python
- **IDE:** VS Code / IDLE
- **Libraries:** Python Standard Library (no external libraries required)
- **Platform:** Windows

• **Implementation**

The program accepts user input in the form of a message and a key.

- For encryption, each character in the message is modified according to the key logic and stored as encrypted text.
- For decryption, the same key is applied in reverse to regenerate the original message. Functions are used to separate encryption and decryption logic, ensuring clarity and modularity of the code.

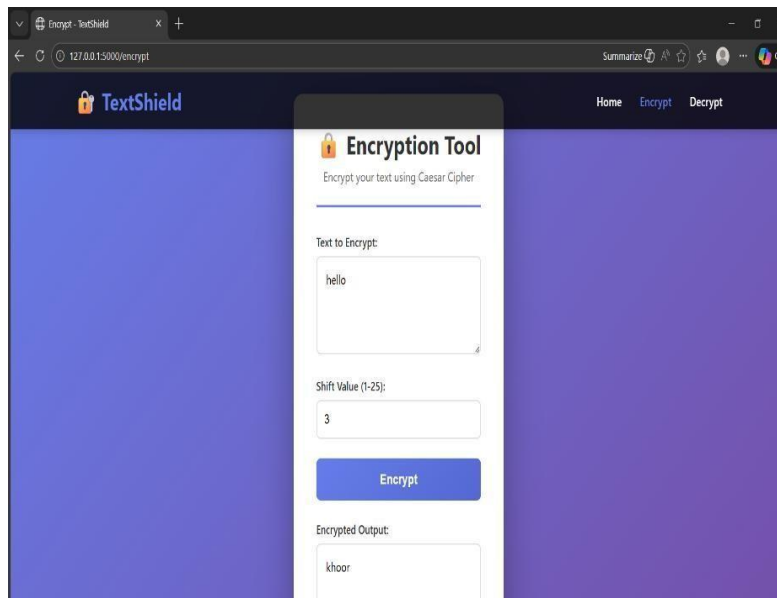
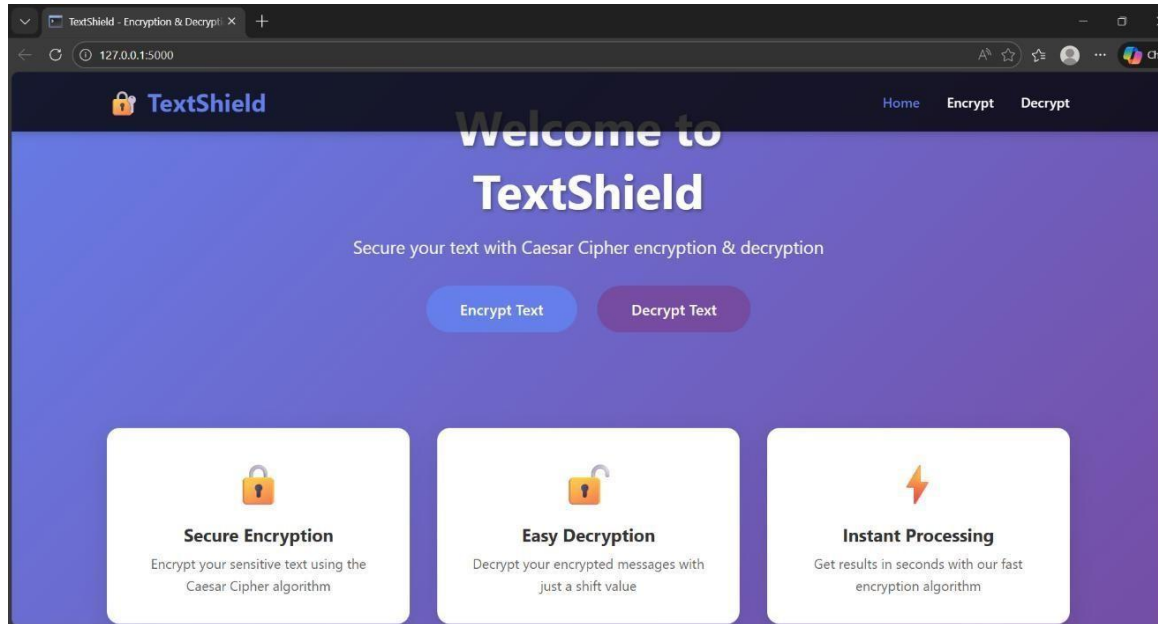
• Code Snippet

•

```
app.py X
app.py > ...
1  from flask import Flask, request, jsonify, render_template
2
3  app = Flask(__name__, template_folder='templates')
4
5  def caesar_cipher(text, shift, mode):
6      result = ""
7      if mode == "decrypt":
8          shift = -shift
9
10     for char in text:
11         if char.isalpha():
12             base = ord('A') if char.isupper() else ord('a')
13             result += chr((ord(char) - base + shift) % 26 + base)
14         else:
15             result += char
16
17     return result
18
19
20 @app.route('/')
21 def home():
22     return render_template("home.html")
23
24
25 @app.route('/encrypt')
26 def encrypt_page():
27     return render_template("encrypt.html")
28
29
30 @app.route('/decrypt')
31 def decrypt_page():
32     return render_template("decrypt.html")
```

```
app.py X
app.py > ...
33
34
35 @app.route('/process', methods=['POST'])
36 def process_text():
37     data = request.json
38     text = data.get("text")
39     shift = int(data.get("shift"))
40     mode = data.get("mode")
41
42     output = caesar_cipher(text, shift, mode)
43     return jsonify({"result": output})
44
45
46 if __name__ == "__main__":
47     app.run(debug=True, port=5000)
```

• Output / Screenshots



• **Result and Conclusion**

The project successfully demonstrates a working encryption and decryption system using Python. It helps in understanding basic data security principles, string manipulation, and logical thinking. This project is suitable for beginners and can be extended further using advanced algorithms like AES or RSA for real-world applications.

MERN STACK

1. Introduction

In today's software industry, full-stack development has become an essential skill for building scalable and dynamic applications. The MERN Stack is one of the most widely used technology stacks due to its efficiency, flexibility, and use of JavaScript across both frontend and backend.

2. Overview of MERN Stack Technologies

2.1 MongoDB

MongoDB is a NoSQL database that stores data in a flexible, JSON-like document format. During the training, the following concepts were covered:

- Introduction to NoSQL databases
- CRUD operations
- Data modeling using Mongoose

2.2 **Express.js**

Express.js is a backend framework built on Node.js that simplifies server-side development. The training included:

- Creating Express servers
- Routing mechanisms
- Middleware implementation

2.3 **React.js**

React.js is a JavaScript library used to build interactive and dynamic user interfaces. Key learnings included:

- Component-based development
- JSX syntax and virtual DOM
- State and props management
- React Hooks (useState, useEffect)

2.4 **Node.js**

Node.js enables JavaScript execution on the server side. During the training, I learned:

- Node.js fundamentals
- Event-driven and asynchronous programming
- Module handling

3. Tools and Technologies Used

- Programming Language: JavaScript (ES6+)
- Frontend: React.js, HTML, CSS
- Backend: Node.js, Express.js
- Database: MongoDB
- Tools: VS Code, Postman, Git/GitHub
- API Communication: Axios / Fetch

4. Work Done During Training

The training involved continuous **hands-on practice and project-based learning**. The major tasks performed during the program include:

- Designing user interfaces using React components
- Implementing backend APIs using Express.js
- Performing CRUD operations with MongoDB
- Integrating frontend and backend using REST APIs
- Implementing basic authentication mechanisms
- Testing APIs using Postman
- Debugging and optimizing application performance

5. Outcome of the Training

The MERN Stack training significantly enhanced my technical capabilities. After completing the program:

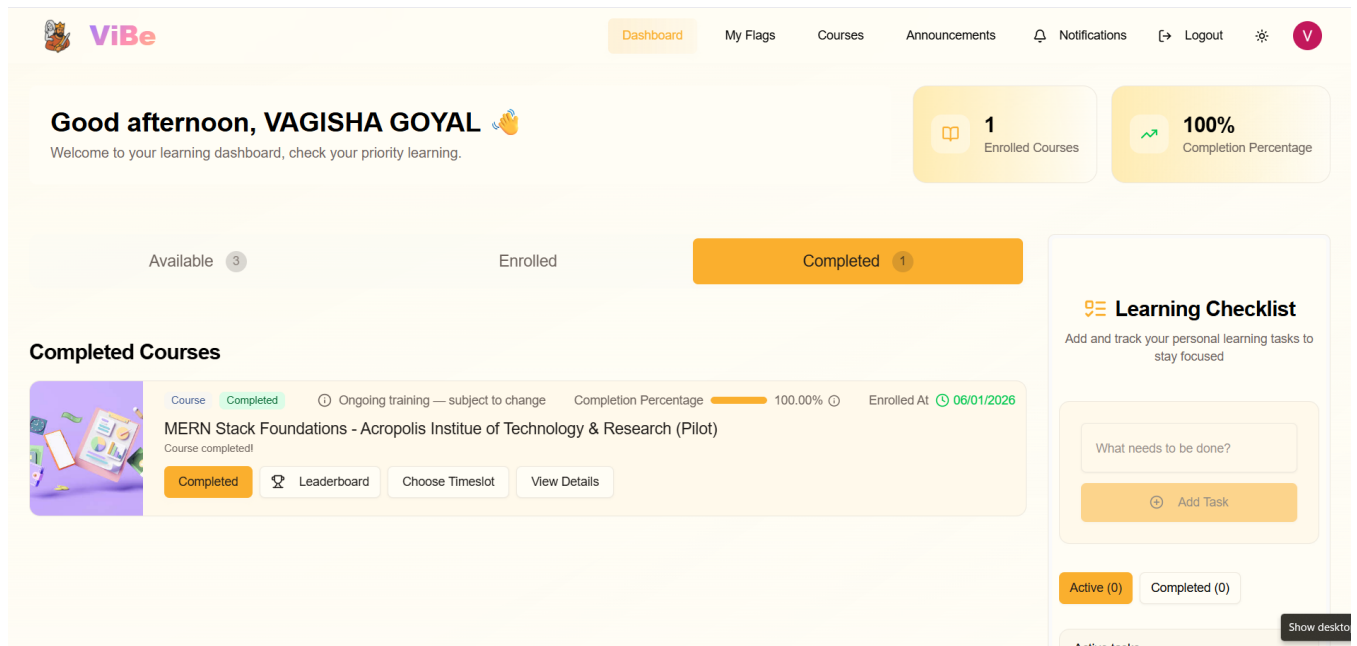
- I can independently develop full-stack web applications
- I gained confidence in working with modern web technologies
- I developed a strong foundation for software development roles
- The certification strengthened my professional and academic profile

6. Conclusion

The MERN Stack training was a valuable and enriching experience that provided comprehensive knowledge of full-stack web development. The combination of theory and hands-on practice helped me understand the complete development lifecycle of web applications.

This training has prepared me for future challenges in the software development field and laid a strong foundation for pursuing a career as a **Full Stack Developer**.

7. Screenshot



IBM CERTIFICATION

Introduction

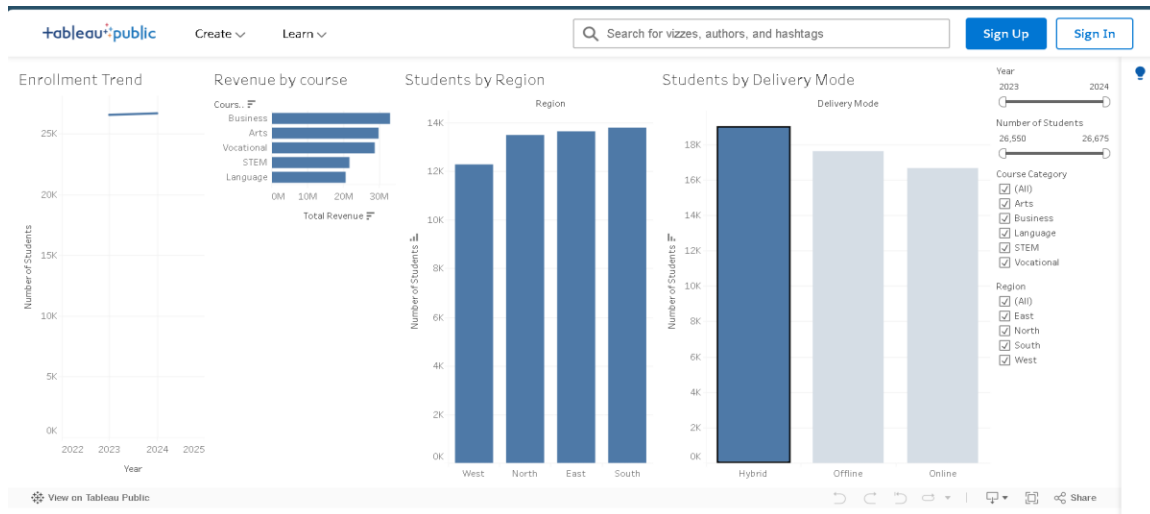
- Data analysis is the process of inspecting, cleaning, transforming, and modeling data to discover useful information.
- In education, it helps identify gaps in access, performance, and equity, supporting SDG 4 (Quality Education).

Tools and Technology Used

Data Collection: Surveys, Government Databases
Processing: Python (Pandas, NumPy), Excel
Visualization: Tableau, Matplotlib, Seaborn
Storage: SQL Databases, Cloud Storage

Implementation

Data Gathering from reliable sources
Cleaning and Transformation of raw data
Visualization using Tableau Dashboards
Statistical Analysis for insights



Results

Access Indicators: Enrollment rates, Literacy levels
 Performance Indicators: Completion rates
 Equity Indicators: Gender parity, Socio-economic gaps
 Visualization Outcome: Interactive Dashboards

Learning

Importance of Clean Data
 Power of Visualization
 Focus on Equity and Inclusion
 Iterative Nature of Data Analysis

Certificate



NPTEL/ GLOBAL Certification /
Hackathons/ Coding Competition

NPTEL:



NPTEL ONLINE CERTIFICATION

(Funded by the MoE, Govt. of India)

This certificate is awarded to
VAGISHA GOYAL
for successfully completing the course

Problem Solving through Programming in C

with a consolidated score of **49** %

Online Assignments	11.88/25	Proctored Exam	36.75/75
--------------------	----------	----------------	----------

Total number of candidates certified in this course: **1641**

Jul-Oct 2025
(12 week course)



Indian Institute of Technology Kharagpur

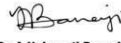


FREE ONLINE EDUCATION
swayam
समग्र शिक्षा, समग्र विकास



Skill India
कौशल भारत - कुशल भारत





Prof. Haimanti Banerji
Coordinator, NPTEL
IIT Kharagpur

Roll No: NPTEL25CS140S1058402185 To verify the certificate  No. of credits recommended: 3 or 4

Forage:





Inspiring and empowering
future professionals

Vagisha Goyal

Cybersecurity Analyst Job Simulation

Certificate of Completion
September 22nd, 2024

Over the period of September 2024, Vagisha Goyal has completed practical tasks in:

- Identity and access management (IAM) fundamentals
- IAM strategy assessment
- Crafting custom IAM solutions
- Platform integration



Tom Brunskill
CEO, Co-Founder of
Forage

Enrollment Verification Code: jxvB9b0dW8iCa7ZQ | User Verification Code: b7f6NgpRfCdcMYqW | Issued by Forage

NSSAFE:



Indian Space Lab:

